PERSONAL EXPENSE TRACKER USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

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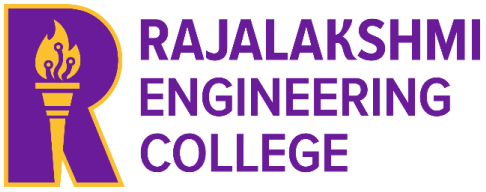
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled **PERSONAL EXPENSE TRACKER** is the bonafide work of **N S ANJAN ADITHYA (2116241401011)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, **N S ANJAN ADITHYA (2116241401011)** hereby declare that the project report titled **PERSONAL EXPENSE TRACKER** submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai **SIGNATURE OF THE STUDENT**

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

ANJAN ADITHYA

ABSTRACT

Managing daily expenses is important for maintaining financial discipline. However, many individuals track their spending using notebooks or spreadsheets, making it difficult to maintain accurate records, calculate total expenses, and understand spending patterns. Manual calculations can lead to errors, consume time, and make expense tracking inconvenient. Therefore, a simple and efficient system is needed to record, organise, and monitor daily expenses.

The Personal Expense Tracker was developed using Zoho Creator, a low-code application development platform. The application allows users to record expenses by entering the expense name, amount, category, and date. It includes an expense entry form, an expense history report, and a dashboard to view recorded expenses and total spending. Deluge scripting and workflow automation are used to update category budgets automatically, and an email notification is sent to the user each time a new expense is added. The application organises expense records in a centralised database and provides a user-friendly interface for accessing information. Access permissions can be configured to protect expense records and restrict unauthorised access.

The application was tested by entering sample expense records, checking the accuracy of stored information, and verifying the expense totals displayed in reports. Testing helps ensure that records are saved correctly and that calculations produce the expected results. The system reduces the need for manual calculations and spreadsheet maintenance while making expense information easier to access and review. Its simple structure also allows future enhancements, such as monthly spending summaries and automated budget alerts. Overall, the Personal Expense Tracker demonstrates how Zoho Creator can be used to develop a practical application with minimal coding effort, improving the convenience and organisation of everyday expense management.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Expense Management; Financial Tracking; Email Notification.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Personal Expense Tracker
Application Domain	Finance – Personal Expense Management
Student Name & Register No.	N S ANJAN ADITHYA – 2116241401011
Zoho Creator Edition	Trial Edition
Application Name (Link Name)	Personal Expense Tracker
Published / Live URL	https://anjanadithya06.zohocreatorportal.in
Demo Credentials (per role)	Admin: Your admin login credentials
Number of Forms / Subforms	2 / 0 (Category Master, Expense Entry)
Number of Reports (by type)	List: 2 (Category Masters, Expense Entries); Kanban: 0; Calendar: 0; Pivot/Chart: 0; Others: 0
Number of Pages / Dashboards	1 Dashboard (Expense Dashboard with 4 summary cards and 2 charts)
Number of Workflows	Form: 1 (Update_Category_Budget); Schedule: 0; Approval: 0; Blueprint: 0; Batch: 0; plus 1 historical recalculation process (DS-02); plus 1 email notification sent on expense submission
Deluge Functions / Custom Actions	2 scripts: DS-01 (budget update), DS-02 (historical recalculation)
Integrations	Zoho Creator built-in database and reporting; email notification through the Zoho Creator mail service
Roles & Profiles Defined	Administrator
Environments Used	Development, Stage and Production
Date of Production Deployment	01 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Expense management is an important part of personal financial planning. Students, employees, and individuals spend money daily on food, transportation, shopping, education, and other activities. Maintaining a record of these expenses helps individuals understand their spending habits and manage their money effectively.

Traditionally, people record expenses in notebooks or spreadsheets. However, these methods require manual data entry and calculations, making it difficult to maintain records regularly. Finding previous transactions and calculating total expenses can also be time-consuming.

The Personal Expense Tracker is developed using Zoho Creator, a low-code application development platform. It provides a simple way to record, organise, and monitor daily expenses. Users can enter expense details, categorise transactions, view expense history, and calculate their total spending. Reports and dashboards help users understand their expenses more clearly.

The application uses Zoho Creator forms, reports, and workflows to simplify expense management. It reduces dependence on manual calculations and provides a centralised location for storing expense records. This project demonstrates how low-code technology can be used to develop a practical application with minimal programming effort.

1.2 Problem Statement

Managing daily expenses manually can make it difficult to maintain accurate financial records. Individuals who use notebooks or spreadsheets may face calculation errors, difficulty organising transactions, and delays in understanding their spending patterns. Manual methods also make it inconvenient to retrieve previous expense details and prepare spending summaries. Therefore, a simple digital application is required to record expenses, organise transactions, and calculate total spending efficiently. The key problems are:

- Manual calculations
- Poor organisation

- Time consumption

1.3 Objectives

- To develop a user-friendly expense tracking application using Zoho Creator.
- To enable users to record expenses with names, amounts, categories, and dates.
- To organise expense records in a centralised database.
- To manage expense categories with allocated budgets and automatically update category-wise spending and remaining balances.
- To provide a dashboard with summary cards and charts for monitoring spending.
- To notify the user by email whenever a new expense is recorded.

1.4 Scope of the Project

In-scope:

- Recording expenses with name, amount, category, and date.
- Viewing and managing expense records.
- Generating expense history reports.
- Calculating and displaying total expenses.
- Providing a dashboard to summarise spending.
- Managing expense categories and allocating category budgets.
- Automatically updating category-wise spending and remaining balances.
- Sending an automatic email notification when a new expense is added.

Out-of-scope:

- Direct integration with bank accounts.
- Online payments and money transfers.
- Automatic detection of transactions from SMS messages.
- Advanced financial forecasting and investment advice.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for developing the Personal Expense Tracker because it provides a low-code environment that reduces the need for extensive programming. Its drag-and-drop form builder allows expense entry forms to be created quickly, while reports and dashboards help organise and display financial information.

The platform supports workflow automation and Deluge scripting, which can be used to perform calculations and automate repetitive tasks. Its integrated data storage simplifies the management of expense records. Zoho Creator also supports web and mobile access, allowing users to manage their expenses conveniently. Built-in access permissions help control access to application data.

Compared with developing an application entirely from scratch, the low-code approach can reduce development time and maintenance effort. It is therefore appropriate for a small project that requires basic data entry, calculations, reports, and automation.

Reference: Zoho Corporation, Zoho Creator Documentation. Available at: https://www.zoho.com/creator/help

1.6 Organisation of the Report

Chapter 1 introduces the project, its background, problem statement, objectives, scope, and justification for using a low-code platform. Chapter 2 reviews existing systems and related literature. Chapter 3 presents the requirement analysis. Chapter 4 explains the system design. Chapter 5 describes the implementation using Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, and security. Chapter 8 presents the results and discussion. Chapter 9 concludes the report and suggests future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

Traditionally, individuals manage their daily expenses using notebooks, calculators, or spreadsheet applications. Every expense is manually recorded along with its amount, date, and purpose. At the end of a day or month, the individual calculates the total spending and reviews the records to understand their expenses.

The existing process involves recording transactions, categorising expenses, calculating totals, and preparing summaries manually. As the number of transactions increases, maintaining accurate records becomes difficult. Manual methods also make it challenging to retrieve previous expenses and analyse spending patterns.

The proposed application replaces this process with a digital system developed using Zoho Creator.

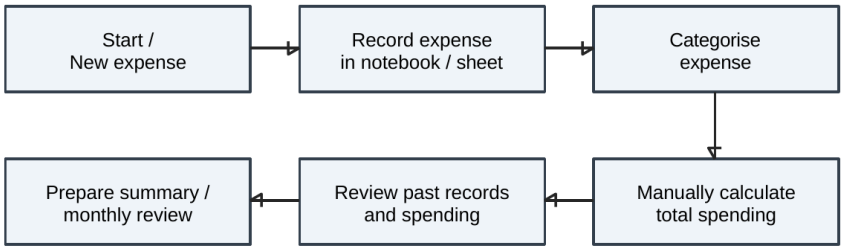


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Manual calculations
- Lack of automation
- Time-consuming reports

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks	Approximately 2–3 weeks
Skills required	Programming and database knowledge	Basic technical knowledge

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Hosting & infrastructure	Must be configured	Platform-managed hosting
Mobile application	Requires additional development	Supported mobile access
Security & access control	Must be implemented	Built-in access permissions
Maintenance & change	Requires code modifications	Can be updated through visual tools
Customisation limits	High flexibility	Depends on platform capabilities
Cost	May require more development resources	Can reduce development effort; subscription charges may apply

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Richardson (2014)	Low-Code Platforms	Faster development	Saves time
2	Gartner (2021)	Low-Code Applications	Simplifies development	Easy building
3	Zoho Creator Documentation	Forms and reports	App development	—
4	Zoho Deluge Documentation	Workflow automation	Auto calculations	—
5	OWASP Security Standards	Data protection	Secure access	—

2.5 Emerging Trends in Low-Code Development

Low-code development is becoming increasingly useful for creating business applications quickly. Several emerging trends are relevant to expense management systems.

1. **AI-Assisted Development:** AI tools can assist developers in creating application components and generating suggestions for improvements.
2. **Workflow Automation:** Automated workflows can calculate totals, validate information, and reduce repetitive manual tasks.

3. **Mobile Application Access:** Mobile support enables users to record and review expenses conveniently from supported devices.
4. **Data Visualisation:** Charts and dashboards help users understand spending patterns and compare expense categories.
5. **Integration Capabilities:** Integrations with other applications and services can support additional functionality when required.

These trends demonstrate how low-code platforms can simplify application development and improve the usability of personal expense management systems.

2.6 Proposed System

The proposed Personal Expense Tracker is a low-code application developed using Zoho Creator. It provides a centralised system for recording, organising, and reviewing daily expenses.

The application includes an expense entry form containing fields for expense name, amount, category, and date. An expense history report displays the recorded transactions, while a dashboard presents the total expenses. Deluge scripting and workflows automate the category budget calculations, and an email notification confirms every expense that is added.

The proposed system addresses the limitations of manual expense management by reducing calculation effort, improving record organisation, and making expense information easier to access.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner	Manage users, settings and records
User	Expense tracker user	Add expenses and view reports

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to add expenses with mandatory fields.	User	High
FR-02	The system shall calculate total expenses automatically.	System	High
FR-03	The system shall allow users to view expense history.	User	Medium
FR-04	The system shall allow users to edit or delete expense records.	User	High
FR-05	The system shall display total expenses on a dashboard.	User	Medium
FR-06	The system shall allow the administrator to manage application settings.	Admin	Medium
FR-07	The system shall validate expense amounts before saving.	System	High
FR-08	The system shall send an email notification to the user when a new expense is added.	System	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Expense reports should load within 3 seconds under normal conditions.
NFR-02	Security	Users shall access only permitted data.
NFR-03	Usability	The application shall be easy to use on web and mobile.
NFR-04	Availability	The application shall be accessible when Zoho Creator is available.

NFR ID	Category	Requirement
NFR-05	Maintainability	Forms and workflows shall be easy to update.
NFR-06	Auditability	Record changes shall be traceable where audit features are enabled.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not required (no external APIs are used)
Other Zoho services	None required
Mobile	Zoho Creator mobile app (optional)
Client hardware	Intel Core i3 or equivalent, 4 GB RAM, internet connection

3.5 Use Case Model

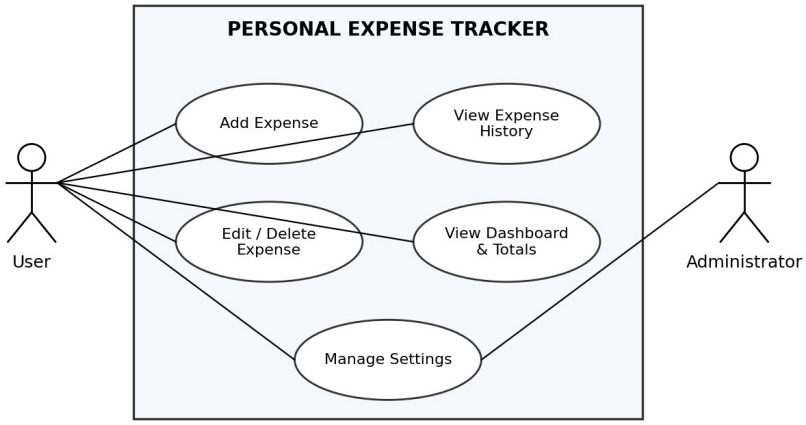


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Add Expense (UC-01)

Use Case ID	UC-01
Use Case Name	Add Expense
Actor(s)	User
Pre-condition	Application is accessible.
Main Flow	1. Open form. 2. Enter details. 3. Submit.

Alternate Flow	Invalid amount → display validation error.
Post-condition	Expense record is saved.
Related FR	FR-01, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

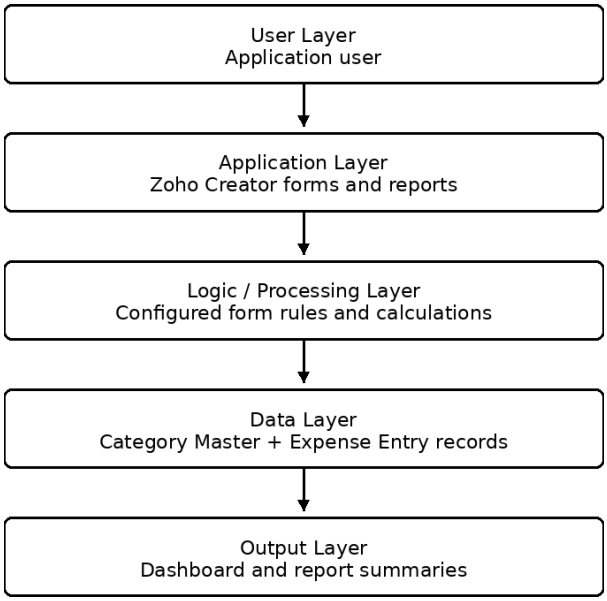


Figure 4.1: System Architecture of the Expense Management System

User Layer: The application user accesses the dashboard, expense entries, and category masters.

Application Layer: Zoho Creator provides the forms, reports, navigation, and dashboard.

Processing Layer: The Update_Category_Budget Deluge workflow runs after each expense is submitted and updates the selected category's Current Spent, Remaining Balance and Last Updated values. A recalculation process (DS-02) synchronises category balances with expenses entered before the workflow existed. An email notification is also sent to the user after each expense is submitted.

Data Layer: Zoho Creator stores the Category Master and Expense Entry records.

Output Layer: The dashboard displays total allocated, current spent, expense-entry count, and category count. An email notification with the expense details is delivered to the user's inbox.

4.2 Data Model

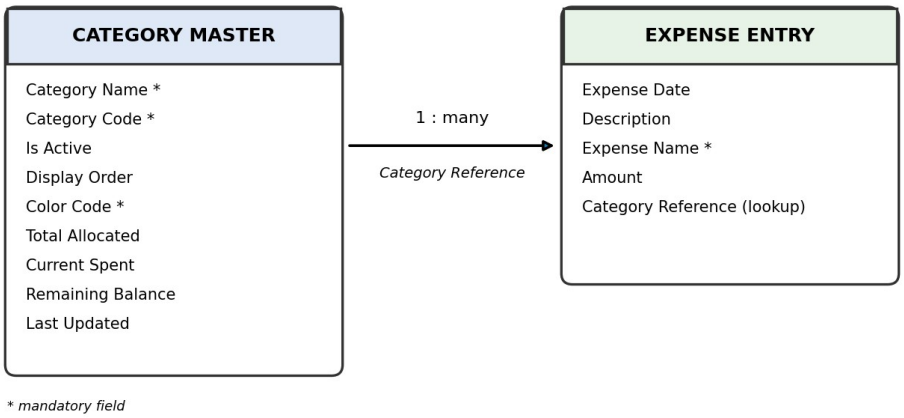


Figure 4.2: Entity–Relationship Diagram of the Expense Management System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Category Master	Category Name, Category Code, Is Active, Display Order, Total Allocated, Current Spent, Remaining Balance, Last Updated	Expense Entry	One-to-Many (via lookup)	Category Reference lookup
Expense Entry	Expense Date, Description, Expense Name, Amount, Category Reference	Category Master	Many-to-One	Lookup field

4.3 Form Design

Table 4.2: Field Specification – Category Master Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Category Name	Category_Name	Single Line	Yes	Required field (marked *)	Identifies category
Category Code	Category_Code	Single Line	Yes	Required field (marked *)	Category reference code
Is Active	Is_Active	Checkbox	No	Checked by default in the form	Enables/disables category flag
Display Order	Display_Order	Number	No	Numeric value	Controls display order
Color Code	Color_Code	Single Line	Yes	Required field (marked *); hex value, e.g. #FF9800	Category display setting

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Total Allocated	Total_Allocated	Currency	No	Currency amount; default 0	Allocated budget
Current Spent	Current_Spent	Currency	No	Currency amount; default 0	Current expenditure
Remaining Balance	Remaining_Balance	Currency	No	Currency amount; default 0	Remaining amount
Last Updated	Last_Updated	Date-Time	No	dd-MMM-yyyy HH:mm:ss format	Update date and time

Table 4.3: Field Specification – Expense Entry Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Expense Date	Expense_Date	Date	No	dd-MMM-yyyy format	Date of the expense
Description	Description	Multi Line	No	Free text	Details of the expense
Expense Name	Expense_Name	Single Line	Yes	Required field (marked *)	Identifies the expense
Amount	Amount	Decimal	No	Numeric, two decimal places	Expense amount
Category Reference	Category_Reference	Lookup	No	Selected from Category Master	Links the expense to a category

4.4 Process and Workflow Design

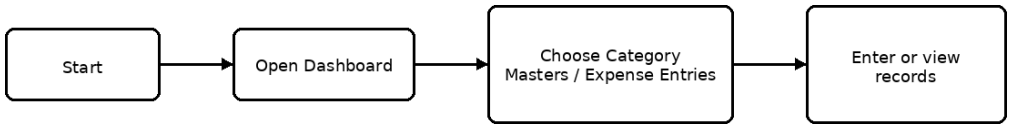


Figure 4.3: Expense Management Process Flow

4.5 User Interface and Navigation Design

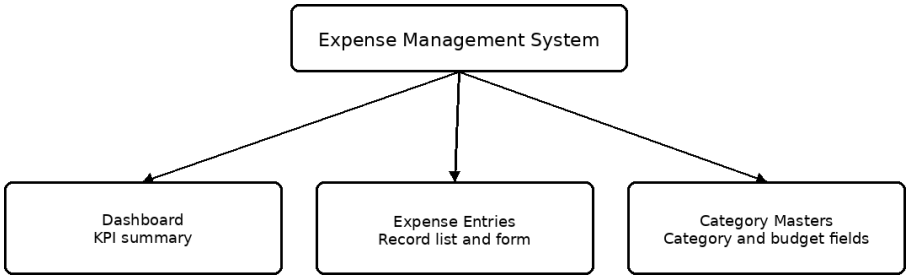


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

In the matrix below, A = Add, E = Edit, V = View and D = Delete. An asterisk (*) means the access scope depends on the permissions configured.

Table 4.4: Role–Permission Matrix

Component	Administrator	Standard User	Portal User	Notes
Category Master form	A E V D	A V	—	Portal not shown/configured
Expense Entry form	A E V D	A E V D*	—	*Access scope depends on permissions
Dashboard	V	V	—	Dashboard visible in screenshots
Expense Entry report	V E D	V E D*	—	*Access scope depends on permissions
Category Master report	V E D	V	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Expense Management System was configured in Zoho Creator as a low-code application. The visible navigation contains Dashboard, Expense Entries, and Category Masters. The dashboard presents summary cards and an expense-record report. Category Masters is used to maintain category details such as category name, code, active status, display order, colour code, allocated amount, current spent amount, remaining balance, and last-updated date. The application uses Zoho Creator forms and reports to store and display the information. The Expense Dashboard also shows two column charts: expenses by category and expenses by month.

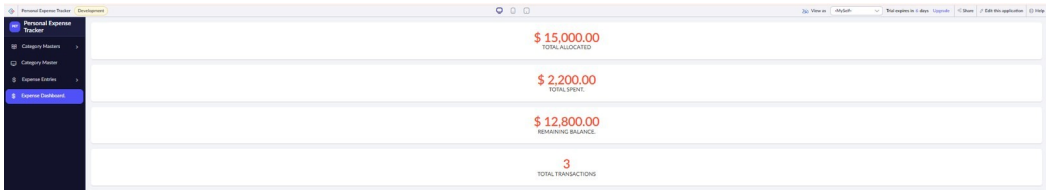


Figure 5.1: Application Dashboard and Navigation Sections

5.2 Forms and Data Capture

The application contains two data-entry forms. The Category Master form captures Category Name, Category Code, Is Active, Display Order, Color Code, Total Allocated, Current Spent, Remaining Balance, and Last Updated. The Expense Entry form captures Expense Date, Description, Expense Name, Amount, and Category Reference. The Category Reference lookup links each expense to its selected category. Both forms provide a Submit control.

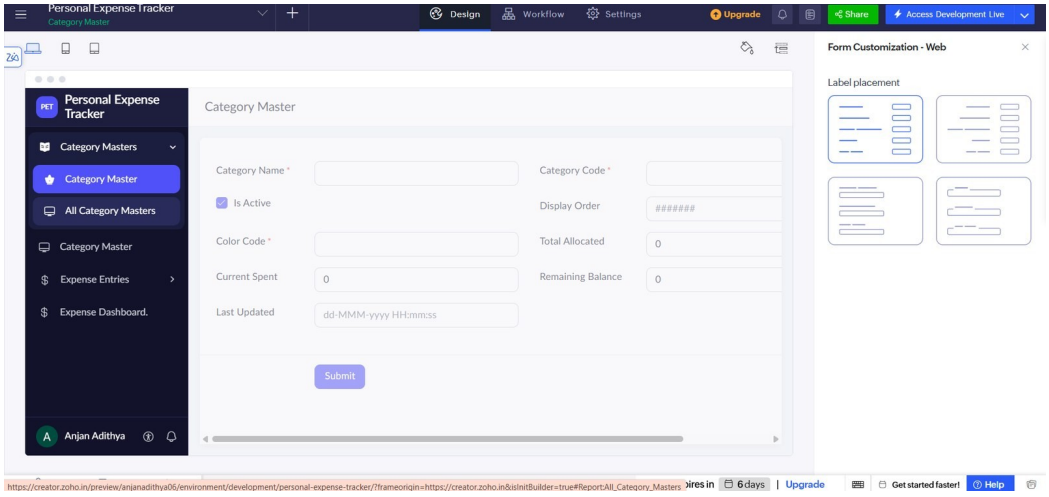


Figure 5.2: Category Master Form

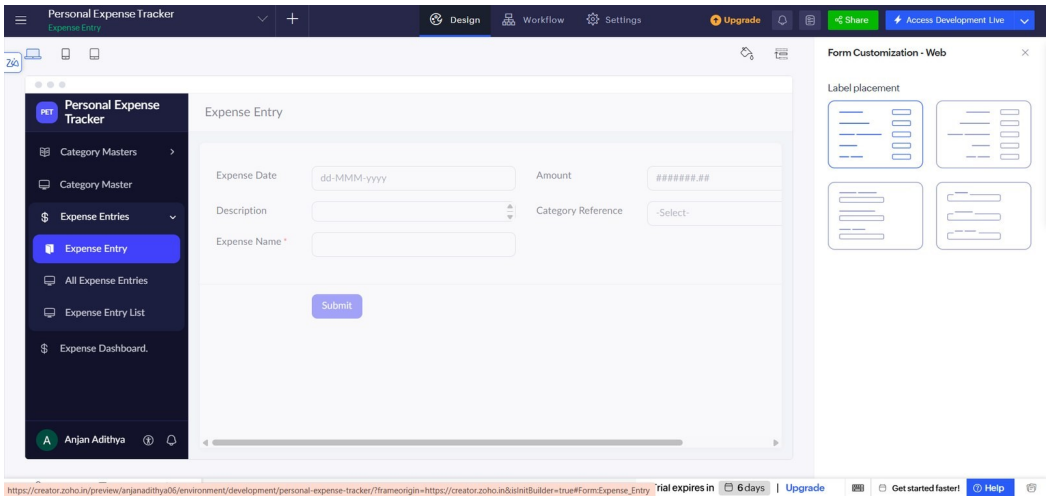


Figure 5.3: Expense Entry Form

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Expense Entries	List report	Expense Entry	Displays expense records; category reference visible	Search / standard report controls	Application user
All Category Masters	List report	Category Master	Displays category name, code, active status, allocation, spent, balance and date	Add record; search; summary view	Application user
Dashboard Summary	Dashboard / KPI cards	Expense Entry and Category Master data	Summary cards for totals and record counts	Category-reference search	Application user

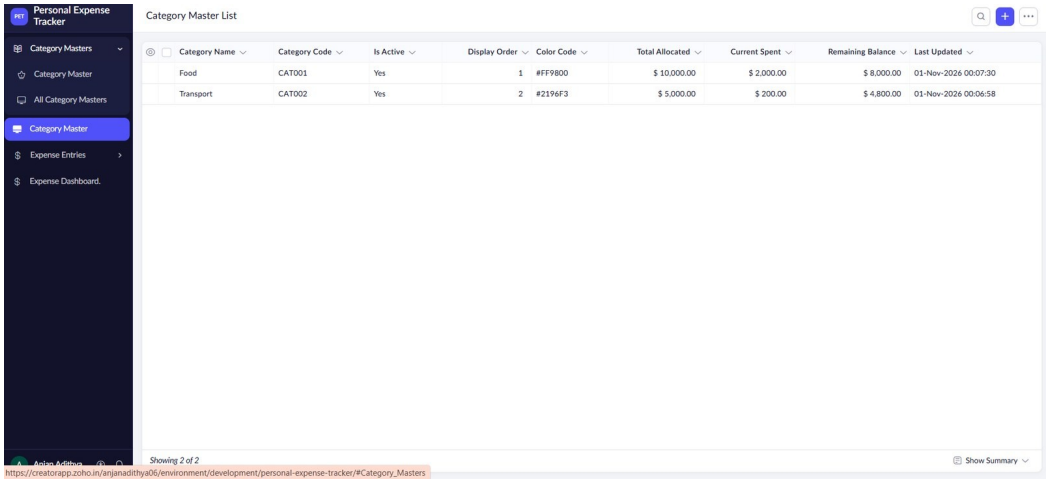


Figure 5.4: Category Masters Report

5.4 Pages and Dashboards

The Expense Dashboard presents Total Allocated, Total Spent, Remaining Balance, and Total Transactions. It also includes a category-wise expense chart and a monthly expense chart. With the sample records entered, the dashboard displays Total Allocated of $15,000.00, Total Spent of $2,200.00, Remaining Balance of $12,800.00, and 3 Total Transactions. The category-wise chart shows Food expenses of $2,000 and Transport expenses of $200; the monthly chart shows $2,200 for October 2026.

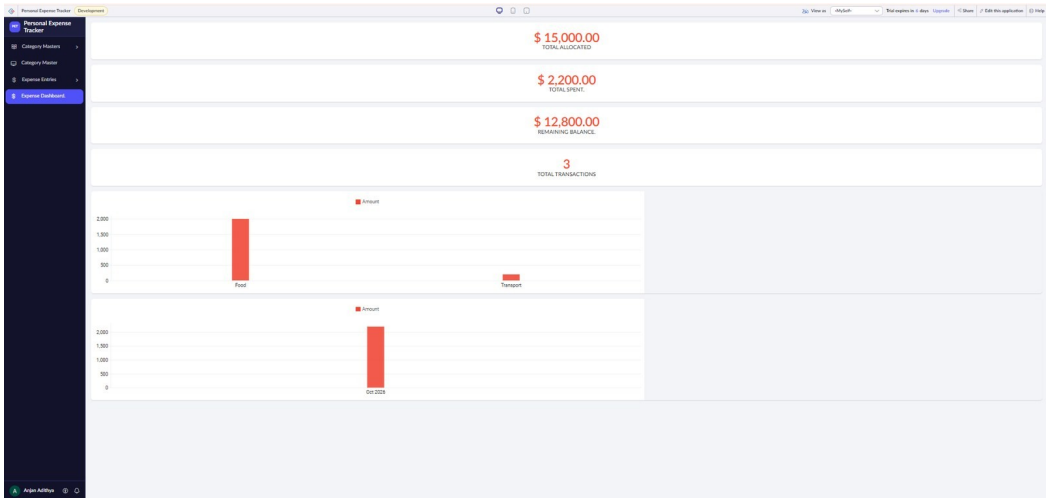


Figure 5.5: Expense Management Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Expense Entry – Created; successful form submission	Updates category spending and remaining balance after an expense is submitted	Form input, lookup, record fetch, conditional logic, arithmetic, record update
DS-02	All Category Masters report – Recalculate Category Budgets action; manual, single-record execution after confirmation	Recalculates the selected category's Current Spent and Remaining Balance from all matching Expense Entry records	Record fetch, summation, record update

A form workflow named Update_Category_Budget is enabled for Expense Entry after successful form submission. Its Deluge script reads the selected Category Reference, retrieves the matching Category Master record, adds the submitted expense amount to Current Spent, recalculates Remaining Balance using Total Allocated, and updates Last Updated. The workflow was verified using the sample expense records shown in the application.

Because Update_Category_Budget runs only when a new expense is submitted, expenses entered before the workflow was created were not counted in the category balances. A recalculation process (DS-02) was therefore configured to calculate each category’s spending from the existing Expense Entry records, so that historical expenses are included in Current Spent and Remaining Balance. No custom validation script has been implemented in the current application.

In addition to the budget update, an email notification is sent after each successful Expense Entry submission. The email, titled “New Expense Added – Personal Expense Tracker”, is sent from notifications@trial.zohocreatormail.com and lists the Expense Name, Expense Date, Amount, Category and Description of the saved record (Figure 5.7).

Code 5.1: Update_Category_Budget – Deluge script of the form workflow (DS-01)

```
// Script ID : DS-01
// Workflow  : Update_Category_Budget
// Trigger   : Expense Entry -> Created -> Successful form submission

categoryRef = input.Category_Reference;
if(categoryRef != null)
{
   categoryRecord = Category_Master[ID == categoryRef];
   if(categoryRecord.count() > 0)
   {
      currentSpent = ifnull(categoryRecord.Current_Spent, 0);
      totalAllocated = ifnull(categoryRecord.Total_Allocated, 0);
      expenseAmount = ifnull(input.Amount, 0);
      newCurrentSpent = currentSpent + expenseAmount;
      newRemainingBalance = totalAllocated - newCurrentSpent;
      categoryRecord.Current_Spent = newCurrentSpent;
      categoryRecord.Remaining_Balance = newRemainingBalance;
      categoryRecord.Last_Updated = zoho.currenttime;
   }
}
```

How the script works:

1. **Identify the category:** input.Category_Reference holds the category selected in the submitted expense; the script continues only if one is selected.
2. **Fetch the Category Master record:** Category_Master[ID == categoryRef] retrieves the matching record, and count() > 0 confirms that it exists.
3. **Read the current values:** Current_Spent, Total_Allocated and the submitted Amount are read with ifnull(…, 0), so an empty value is treated as zero.
4. **Calculate:** the new Current Spent is the old value plus the expense amount, and the new Remaining Balance is Total Allocated minus the new Current Spent.
5. **Update the record:** the new Current_Spent and Remaining_Balance are written back to the category, and Last_Updated is set to the current time (zoho.currenttime).

Code 5.2: Historical recalculation script (DS-02)

```
// Script ID : DS-02
// Purpose   : Recalculate category spending from existing
//             Expense Entry records (historical synchronisation)

categories = Category_Master[ID != 0];
for each cat in categories
{
   totalSpent = 0;
   expenses = Expense_Entry[Category_Reference == cat.ID];
   for each expRec in expenses
   {
      totalSpent = totalSpent + ifnull(expRec.Amount, 0);
   }
   totalAllocated = ifnull(cat.Total_Allocated, 0);
   cat.Current_Spent = totalSpent;
   cat.Remaining_Balance = totalAllocated - totalSpent;
   cat.Last_Updated = zoho.currenttime;
}
```

The script loops through every Category Master record, adds up the Amount of all Expense Entry records that refer to that category, and then rewrites Current Spent, Remaining Balance and Last Updated. Running it after the workflow was created brings the earlier expenses into the balances.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Category entry	Form submission	Submit button	Entered category data	Save category record through form configuration	None verified
Expense entry	Form workflow	Created – successful form submission	Category Reference is selected	Save expense record and update Category Master Current Spent and Remaining Balance	DS-01 – Update_Category_Budget
Expense email notification	Form workflow – email	Created – successful form submission	Expense record is saved	Send the email “New Expense Added – Personal Expense Tracker” with expense name, date, amount, category and description	Fig. 5.7

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
History sync	Recalculation process	All Category Masters report action – selected record; manual execution after confirmation	Existing Expense Entry records for the selected category	Recalculate category Current Spent, Remaining Balance and Last Updated	DS-02
Dashboard summary	Report / page display	Open dashboard	Existing records available	Display summary cards and reports	Built-in display

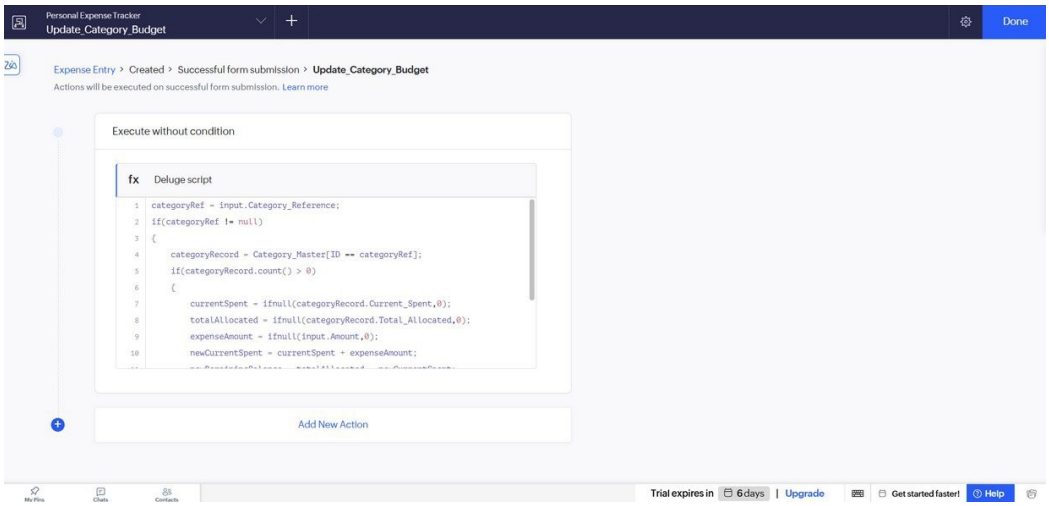


Figure 5.6: Expense Entry Workflow and Deluge Budget Update

The email notification runs when an Expense Entry record is successfully submitted. Figure 5.7 shows the message received in the inbox after the expense LUNCH (Food, 200.00, dated 01-Nov-2026) was submitted. The earlier message in the same thread shows that the first version of the email displayed unreplaced placeholders such as [Insert Expense_Name] instead of the entered values; the second message, sent after the content was corrected, shows the actual expense details (see BUG-03 in Table 6.2).

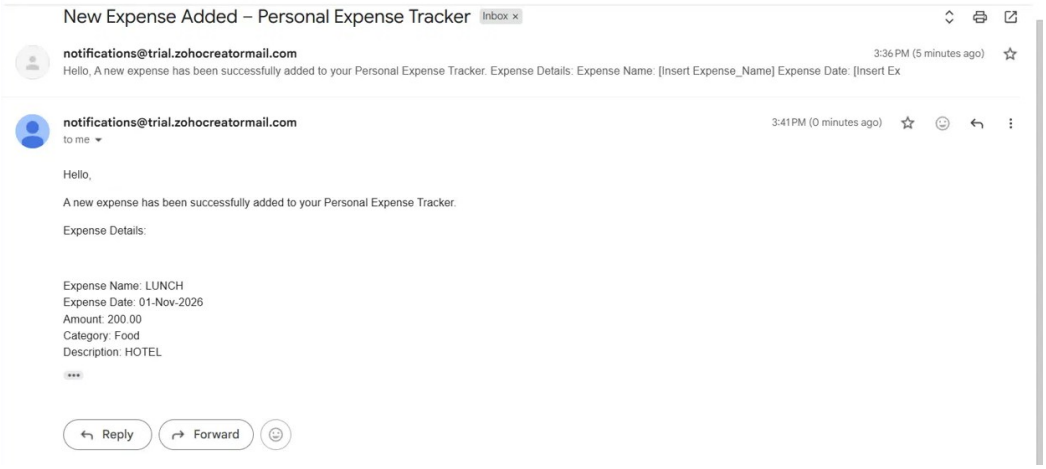


Figure 5.7: Email Notification Received for a Newly Added Expense

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email notification	Zoho Creator built-in email service	Form workflow – send email	Platform-managed	Sends the saved expense details to the user's inbox (Fig. 5.7)

No external API, REST call, webhook or third-party service is used; the email notification is delivered through the platform's built-in email service.

Code 5.3: External integration script (not used)

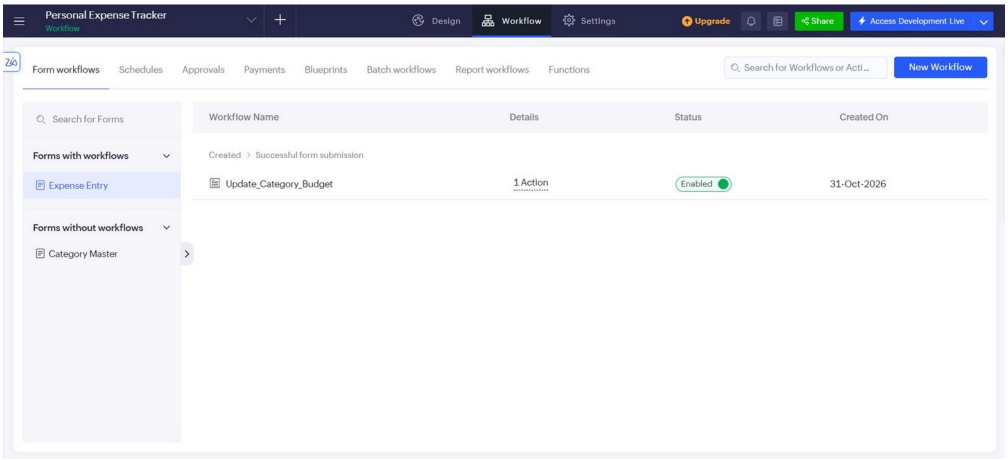


Figure 5.8: Workflow module of Zoho Creator showing the available workflow types; the project uses only a form workflow

5.8 User Management and Role-Based Access

The application screenshot shows the signed-in user interface and the three main navigation sections. The screenshots do not establish that separate permission profiles, record-level rules, or a customer portal have been configured. Access control should be verified in Zoho Creator before claiming role-specific restrictions. The application should be shared only with authorised users and permissions should follow the minimum-access principle.

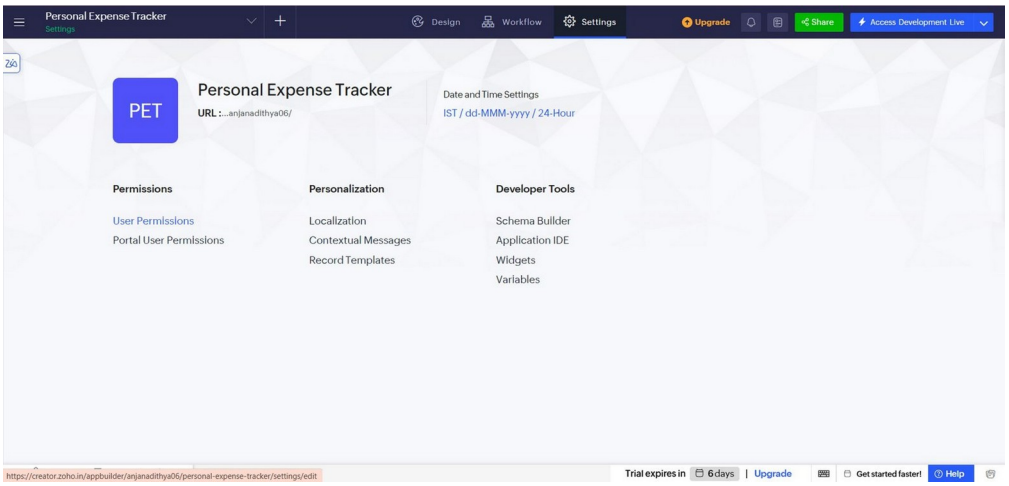


Figure 5.9: Application settings showing User Permissions and Portal User Permissions

5.9 AI and Analytics Features

The Expense Dashboard shows four summary cards (Total Allocated, Total Spent, Remaining Balance and Total Transactions) and two column charts built from the Expense Entry data: expenses by category and expenses by month. No Zia feature, predictive model, OCR, or Zoho Analytics integration is evidenced in the supplied screenshots. The dashboard is therefore described as a visual summary of application data built with Zoho Creator's own page components, not as an AI analytics system. Figure 5.10 shows the dashboard in the page builder, where the cards and charts are configured.

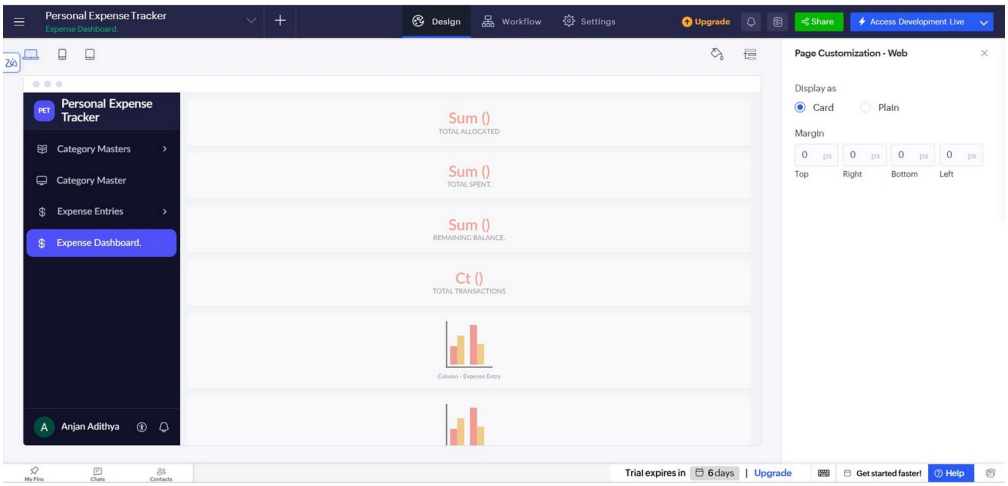


Figure 5.10: Expense Dashboard in the page builder showing summary cards and chart components

5.10 Pro-Code Extension

The current project uses Zoho Creator forms, reports, lookup fields, and dashboard components. No pro-code extension such as Zoho Catalyst, custom JavaScript widgets, or external REST API is shown in the supplied evidence. A pro-code component is not necessary for the current basic expense-recording scope. It could be considered later if the project requires complex integrations or processing beyond the built-in low-code features.

5.11 Mobile and Portal Access

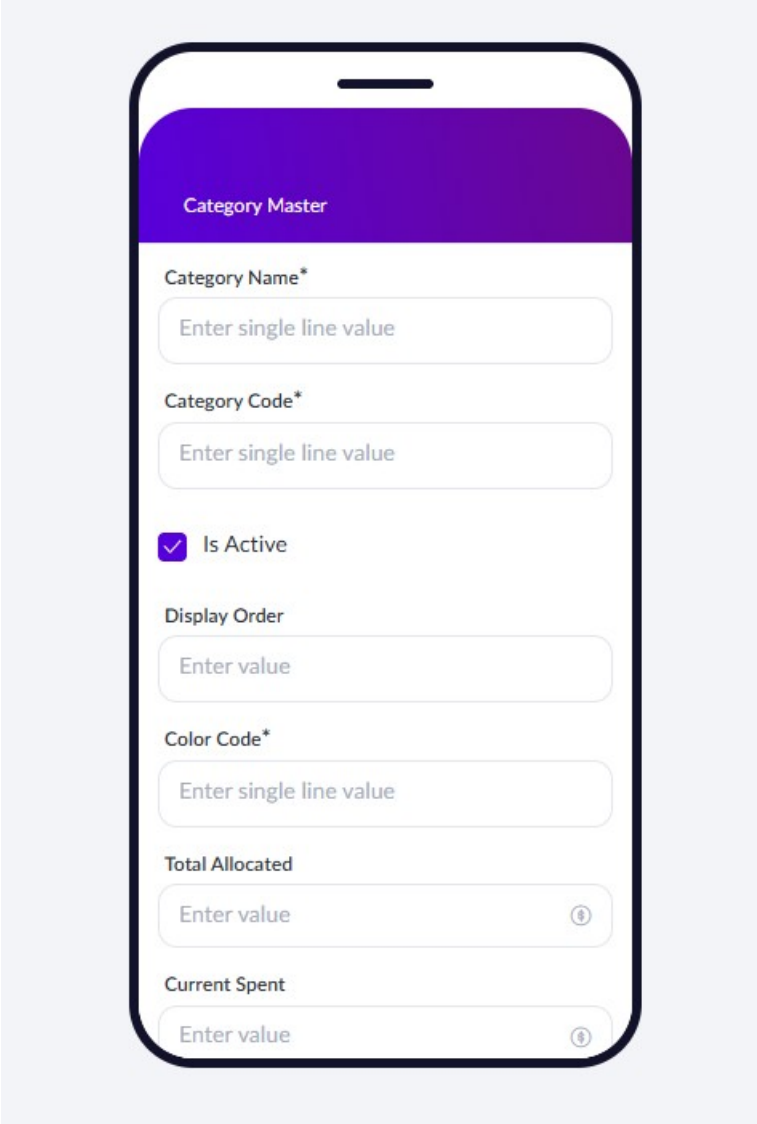


Figure 5.11: Illustrative mobile layout concept

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is organised around the visible application functions: opening each module, entering category details, submitting an expense entry, checking the category lookup, viewing saved records, and reviewing dashboard totals. Negative tests include missing mandatory values, invalid numeric values, and selecting an inactive category if the form is configured to prevent it. Where the supplied screenshots show the outcome, the test table records it as a pass with the figure that supports it. Tests without screenshot evidence are left as pending for confirmation in the live application. The email notification is checked by submitting an expense and confirming that a message with the saved details reaches the inbox.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Open Dashboard	Select Dashboard in navigation	Dashboard cards and report are displayed	Cards and charts displayed (Fig. 5.5)	Pass
TC-02	FR-01	Open Expense Entries	Select Expense Entries	Expense entry report is displayed	Not independently verified	Pending live test
TC-03	FR-01	Open Category Masters	Select Category Masters	Category master report is displayed	Category Master List displayed (Fig. 5.4)	Pass
TC-04	FR-01	Open Category Master form	Click add (+) control	Category form opens	Category form opens (Fig. 5.2)	Pass
TC-05	FR-01	Enter category name	Enter a valid category name	Name is accepted by the form	Not independently verified	Pending live test
TC-06	FR-01	Enter category code	Enter a category code	Code is accepted by the form	Not independently verified	Pending live test
TC-07	FR-01	Set active status	Select Is Active checkbox	Checkbox state can be changed	Not independently verified	Pending live test
TC-08	FR-01	Enter display order	Enter a numeric display order	Numeric value is accepted	Not independently verified	Pending live test
TC-09	FR-01	Enter allocation amount	Enter a valid currency amount	Amount is accepted	Not independently verified	Pending live test
TC-10	FR-01	Submit category form	Complete fields and submit	Category record is saved and shown in report	Food and Transport records saved and listed (Fig. 5.4)	Pass
TC-11	FR-01	Reset category form	Enter values and select Reset	Form fields return to default state	Not independently verified	Pending live test

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-12	FR-02	Open Expense Entry form	Click add (+) in Expense Entries	Expense form opens	Expense form opens (Fig. 5.3)	Pass
TC-13	FR-02	Select category lookup	Choose a Category Reference	Selected category is displayed	Not independently verified	Pending live test
TC-14	FR-02	Submit expense entry	Enter available details and submit	Expense record is saved and listed	Dashboard counts 3 saved transactions (Fig. 5.5)	Pass
TC-15	FR-03	Search category reference	Enter a known reference and search	Matching expense entries are displayed if present	Not independently verified	Pending live test
TC-16	FR-04	View dashboard totals	Open Dashboard after saving sample records	Cards display the current stored summary values	Cards show $15,000.00 / $2,200.00 / $12,800.00 / 3 (Fig. 5.5)	Pass
TC-17	NFR-02	Check unauthorised access	Attempt access with an account without permission	Access is denied if permissions are configured	Not independently verified	Pending live test
TC-18	NFR-01	Check report display	Open reports under normal use	Report loads and displays available records	Not independently verified	Pending live test
TC-19	FR-02	Verify budget update workflow	Submit an expense with a Category Reference selected	Category Current Spent and Remaining Balance are updated	Food: $2,000 spent, $8,000 left; Transport: $200 spent, $4,800 left (Fig. 5.4)	Pass
TC-20	FR-02	Verify historical recalculation	Run the recalculation process on existing Expense Entry records	Category Current Spent and Remaining Balance include all existing expenses	Not independently verified	Pending live test
TC-21	FR-08	Verify email notification	Submit an expense (Expense Name LUNCH, Category Food, Amount 200.00)	An email with the expense details is received	Email received showing LUNCH, 01-Nov-2026, 200.00, Food, HOTEL (Fig. 5.7)	Pass

6.3 Debugging Deluge Scripts

Debugging for this application should begin by checking the form field link names against the names used in any workflow, then reviewing mandatory-field settings and lookup configuration. For Deluge code, the editor’s error message and info statements can help identify invalid field references or incorrect data types. No verified Deluge error log was supplied for this report, so a specific resolved script defect is not claimed.

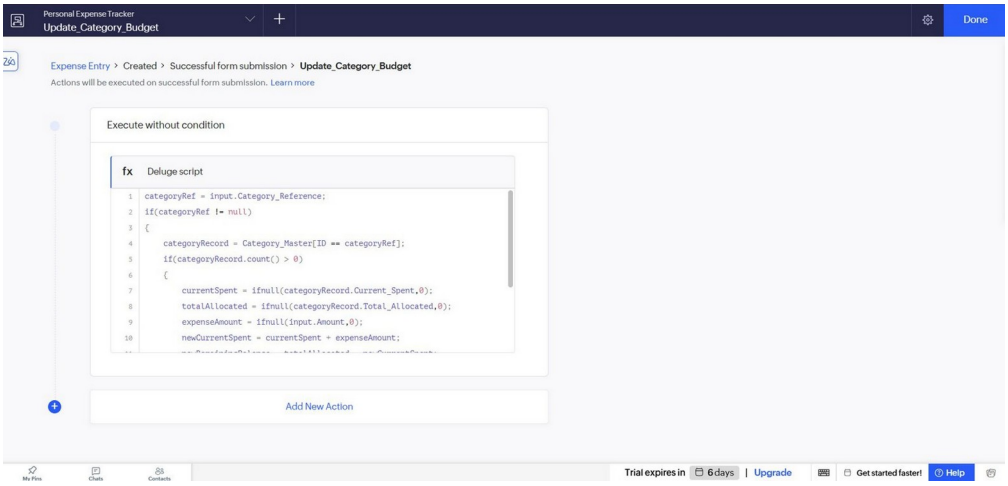


Figure 6.1: Deluge script editor of the Update_Category_Budget workflow

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Category Master report showed blank category details and Is Active as false in the captured view	The supplied screenshots show blank values for some category fields; cause not confirmed	Valid category records (Food, Transport) entered; Is Active shows Yes and all fields display in the report (Fig. 5.4)	Closed
BUG-02	Dashboard totals displayed $0.00 in the captured state	No non-zero allocation or expense values are visible in the screenshot	Sample allocation and expense records entered; dashboard now shows $15,000.00 / $2,200.00 / $12,800.00 (Fig. 5.5)	Closed
BUG-03	First email notification showed unreplaced placeholders such as [Insert Expense_Name] instead of expense values	Message content contained placeholder text instead of the Expense Entry field values	Message content corrected; the next email shows LUNCH, 01-Nov-2026, 200.00, Food, HOTEL (Fig. 5.7)	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Not recorded	Application user	Formal user feedback was not supplied	Collect feedback after live user acceptance testing
Not recorded	Application owner	Formal user feedback was not supplied	Collect feedback after live user acceptance testing

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Zoho Creator Development environment, then moved through the Stage environment before release. Version 1.0 was published to the Production environment on 01 November 2026. The Environments page confirms Version 1.0 in both Stage and Production. The published application URL is https://anjanadithya06.zohocreatorportal.in. Future changes should be validated in Stage before publishing a new version to Production. The application owner should retain a previous version or export for recovery and review access permissions before wider use.

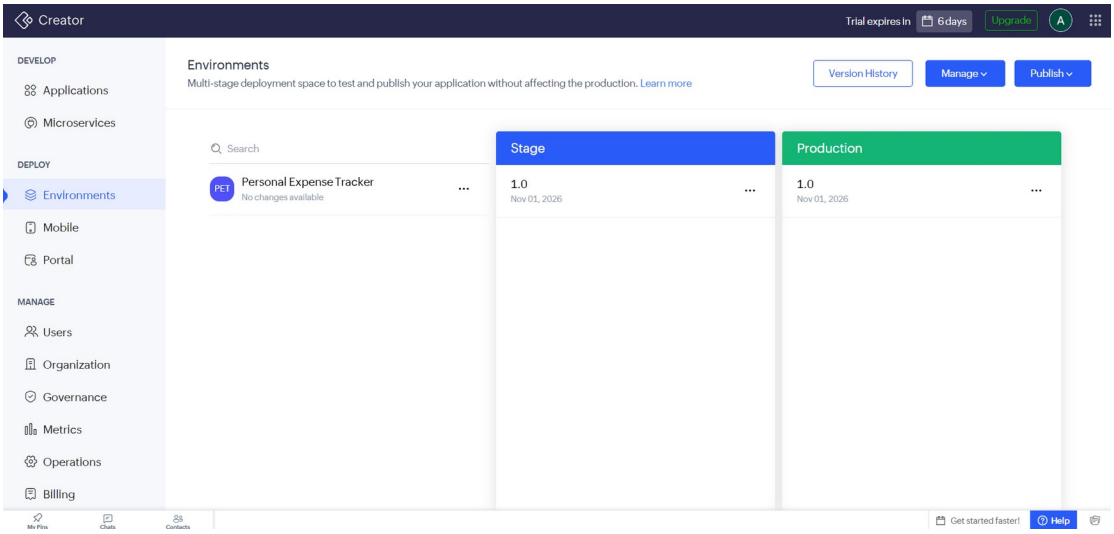


Figure 7.1: Illustrative deployment checklist

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Open all application modules	Yes – screenshots supplied
2	Test Category Master form submission	Yes – records saved (Fig. 5.4)
3	Test Expense Entry form submission	Yes – 3 transactions counted (Fig. 5.5)
4	Verify category lookup	Yes – category totals update (Fig. 5.4)
5	Verify dashboard calculations with sample data	Yes – totals match (Fig. 5.5)
6	Verify email notification for a new expense	Yes – email received (Fig. 5.7)

S.No.	Checklist Item	Done (✓)
7	Verify permissions for different users	Pending live test
8	Export or back up application data	Pending confirmation

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Use filtered reports when reviewing specific records	Expense Entries report	Not measured	Not measured	Reduces the displayed record set when filters are used
Keep report columns relevant to the task	Expense Entries / Category Masters	Not measured	Not measured	Improves readability; timing not measured
Use summary cards for quick overview	Dashboard	Detailed review required	Summary shown on dashboard	Faster visual overview; no timed benchmark available

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho Creator account sign-in	Application access	Unauthorised access to the account
Role/profile permissions	Must be verified in application settings	Unauthorised access to forms and reports
Avoid storing secrets in scripts	No external integrations shown	Exposure of credentials

7.4 Governance and Administration

The application should be maintained by its owner. Recommended practices include using clear form and field names, checking records for accuracy, reviewing user access when responsibilities change, exporting a backup before major modifications, and testing changes before publishing. Audit-trail and version-history settings should be checked in Zoho Creator because the supplied screenshots do not demonstrate their configuration; they should be verified before claiming they are enabled.

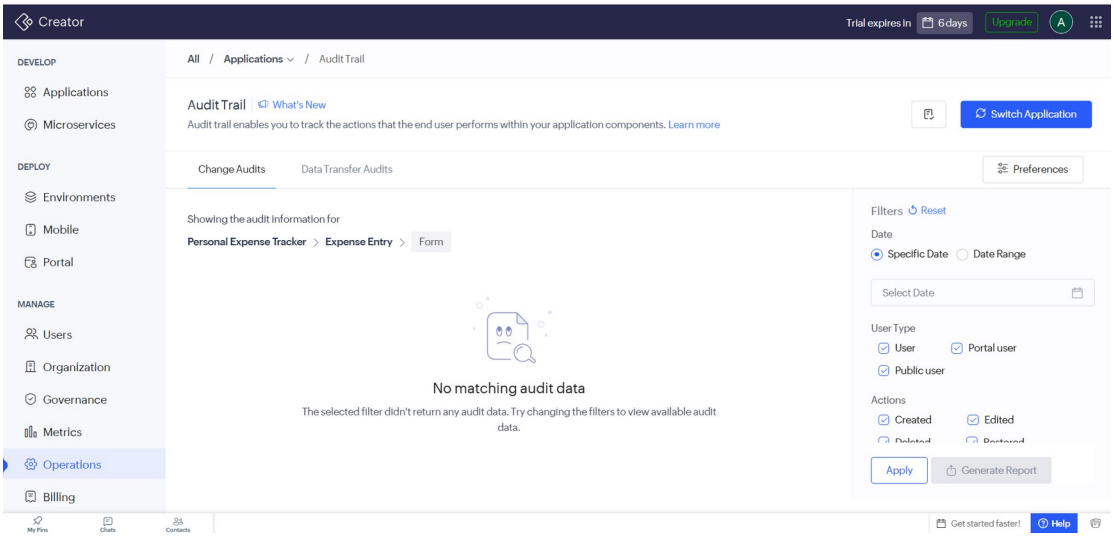


Figure 7.2: Illustrative audit trail and version history

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A typical usage scenario begins when the user opens the Expense Dashboard to review the summary cards and charts. The current dashboard displays Total Allocated of $15,000.00, Total Spent of $2,200.00, Remaining Balance of $12,800.00, and 3 Total Transactions. The user can then open Category Masters to review category budgets and balances. To record a new transaction, the user opens the Expense Entry form, enters the Expense Date, Description, Expense Name, and Amount, selects the relevant Category Reference, and submits the record. The enabled Update_Category_Budget workflow then updates the selected category's Current Spent and Remaining Balance. An email notification with the expense details is then sent to the user's inbox (Figure 8.4).

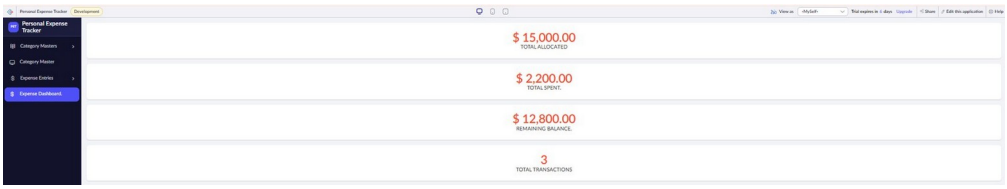


Figure 8.1: Step 1 – Review the Dashboard

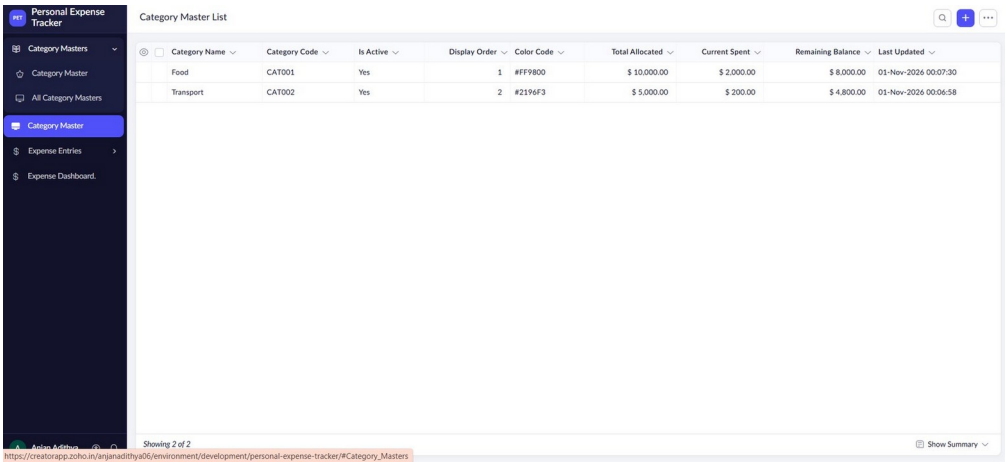


Figure 8.2: Step 2 – Review Category Masters

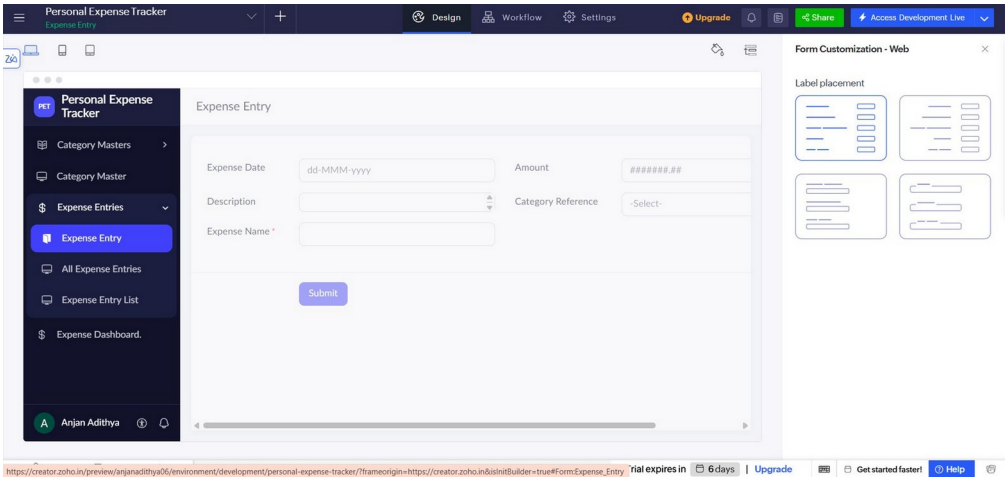


Figure 8.3: Step 3 – Complete and Submit an Expense Entry

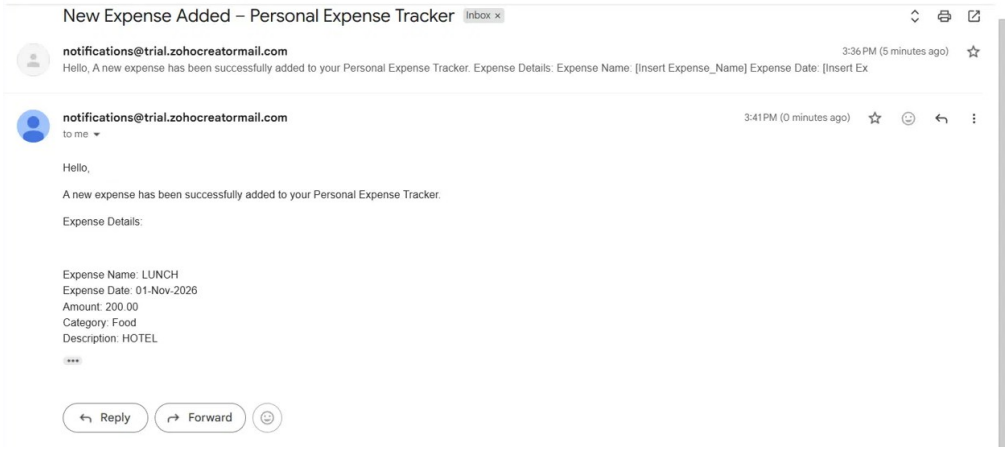


Figure 8.4: Step 4 – Receive the Email Notification

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Centralise expense information	Expense Entries report	Expense records are shown in one report
Maintain expense categories	Category Master form and report	Category fields and report are visible
Automatically update category budgets	Update_Category_Budget workflow (DS-01); recalculation process (DS-02)	Food and Transport balances match the recorded expenses (Fig. 5.4)
Provide a spending overview	Dashboard cards	Cards and charts show allocated, spent, balance and record counts
Make expense information easier to review	Reports and category-reference search	Report screens and search field are visible

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Notify the user of each new expense	Email notification on Expense Entry submission	Email with expense name, date, amount, category and description received (Fig. 5.7)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording an expense	Manual notebook or spreadsheet entry	Zoho Creator Expense Entry form	Centralised digital entry
Category maintenance	Separate manual list	Category Master form and report	Categories are stored in a structured form
Reviewing totals	Manual calculation	Dashboard summary cards	Quick visual summary; totals verified against sample data
Finding records	Manual search in notes or sheets	Reports and search control	More organised record review
Confirmation of an entry	No confirmation; the user checks the notebook or sheet	Automatic email notification	Immediate confirmation of every recorded expense

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator's platform, data model, and available configuration options. The visual layout is easier to build than a fully custom-coded interface but offers less freedom for detailed UI behaviour. Advanced calculations, external integrations, or complex financial analysis may require Deluge scripting or external services. The project also depends on access to the Zoho Creator account and its available plan features. The current scope avoids external integrations and focuses on basic category and expense management.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, lookup field, reports, navigation, dashboard cards and charts represent the no-code configuration side of the spectrum. Deluge scripting is the low-code layer: it is used for the automatic category budget update (DS-01) and the historical recalculation (DS-02), which cannot be achieved through standard field settings. No pro-code extension is used; pro-code would be considered only for requirements such as a specialised external integration or complex processing.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Expense Management System was developed using Zoho Creator to organise expense records and category information in one application. It provides Dashboard, Expense Entries, and Category Masters sections. The Category Master form captures category details and budget-related fields, while the Expense Entry form uses a Category Reference lookup. Reports display stored records, and the dashboard presents summary cards for allocated amounts, current spending, remaining balance, and transaction count. The dashboard also presents category-wise and monthly expense charts, and the Update_Category_Budget workflow keeps each category's Current Spent and Remaining Balance up to date, while a recalculation process (DS-02) brings in expenses entered before the workflow was created. An email notification is also sent to the user each time a new expense is saved. These features provide a centralised alternative to notebooks or separate spreadsheets for basic expense tracking. The sample data confirms that the dashboard totals and category balances are consistent; access restrictions and deployment settings should still be verified in the live application before being treated as completed features.

9.2 Future Enhancements

1. Extend the existing email notification into budget-limit alerts that warn the user when a category's Remaining Balance falls below a set threshold.
2. Extend the budget update so that Current Spent and Remaining Balance are also corrected when an expense is edited or deleted (the current workflow runs only when an expense is created).
3. Add date filters and downloadable monthly expense summaries; consider external integrations only if required.

9.3 Learning Outcomes and Career Relevance

This project provides practical experience in designing forms, linking related records, organising reports, and presenting data through a dashboard using a low-code platform. It also highlights the importance of field naming, data validation, testing, and access control. These skills are relevant to low-code development, business process

automation, application support, and solutions consulting. Further practice with Deluge, workflow testing, and secure deployment would help extend the application beyond its current scope.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Dashboard, Expense Entries, Category Masters	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Yes	Category Master and Expense Entry forms	5.2	Fig. 5.2, 5.3
II	Lookup field (relationship between forms)	Yes	Expense Entry → Category Reference	4.2, 5.2	Fig. 5.3
II	Field validations (mandatory fields, data types, default values)	Yes	Both forms	4.3, 5.2	Table 4.2, 4.3
II	List reports with search	Yes	All Expense Entries; All Category Masters	5.3	Fig. 5.4
II	Page / dashboard built with UI Builder	Yes	Expense Dashboard	5.4	Fig. 5.5
II	Deluge – form event (on successful submission)	Yes	Expense Entry – Created	5.5	Code 5.1
II	Deluge – record fetch, loops and arithmetic	Yes	DS-01, DS-02	5.5	Code 5.1, 5.2
II	Email notification	Yes	Expense Entry – on successful submission	5.5, 5.6	Fig. 5.7
II	Form workflow	Yes	Update_Category_Budget	5.6	Fig. 5.6
II	Testing & debugging evidence	Yes	Test cases and defect log	6	Fig. 6.1
III	Analytics (dashboard charts)	Yes	Expense Dashboard	5.4, 5.9	Fig. 5.5, 5.10
V	Low-code limitations & spectrum analysis	Yes	Report analysis	8.4, 8.5	–

Total features implemented: 13

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Satisfactory	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Satisfactory	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Satisfactory	

Self-assessment scale: Excellent, Good, Satisfactory or Needs Improvement. The Examiner Level column is left blank for the examiner. Basis for the self-assessment:

- **CO1 – Good:** the principles, architecture and justification for low-code development are covered in Chapters 1 and 2.
- **CO2 – Good:** two forms with a lookup, list reports, a dashboard, a form workflow, Deluge scripts (DS-01, DS-02) and an email notification were built and tested.
- **CO3 – Satisfactory:** the only outbound feature is the email notification; no API, connection or external system is used, and role-based access was not independently verified.
- **CO4 – Satisfactory: the application was tested in Development and Stage, and Version 1.0 was published to Production on 01-Nov-2026. Detailed role permissions and audit controls still require verification.**
- **CO5 – Satisfactory:** the limitations of low-code and the no-code to pro-code spectrum are analysed; no AI-assisted or pro-code component was used.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Update_Category_Budget (DS-01)

```
// Script ID : DS-01
// Location  : Expense Entry → Created → Successful form submission
// Purpose   : Update category Current Spent, Remaining Balance and Last Updated
categoryRef = input.Category_Reference;
if(categoryRef != null)
{
   categoryRecord = Category_Master[ID == categoryRef];
   if(categoryRecord.count() > 0)
   {
      currentSpent = ifnull(categoryRecord.Current_Spent, 0);
      totalAllocated = ifnull(categoryRecord.Total_Allocated, 0);
      expenseAmount = ifnull(input.Amount, 0);
      newCurrentSpent = currentSpent + expenseAmount;
      newRemainingBalance = totalAllocated - newCurrentSpent;
      categoryRecord.Current_Spent = newCurrentSpent;
      categoryRecord.Remaining_Balance = newRemainingBalance;
      categoryRecord.Last_Updated = zoho.currenttime;
   }
}
```

C.2 Historical recalculation (DS-02)

```
// Script ID : DS-02
// Location  : All Category Masters report action – Recalculate Category Budgets
// Purpose   : Recalculate category spending from existing Expense Entry records
categoryRecord = Category_Master[ID == input.ID];
if(categoryRecord.count() > 0)
{
   totalSpent = 0;
   expenses = Expense_Entry[Category_Reference == categoryRecord.ID];
   for each expRec in expenses
   {
      totalSpent = totalSpent + ifnull(expRec.Amount, 0);
   }
   totalAllocated = ifnull(categoryRecord.Total_Allocated, 0);
   categoryRecord.Current_Spent = totalSpent;
   categoryRecord.Remaining_Balance = totalAllocated - totalSpent;
   categoryRecord.Last_Updated = zoho.currenttime;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

The Personal Expense Tracker is a Zoho Creator application. It is used through a web browser and can also be opened from the Zoho Creator mobile app.

1. **Open the application.** Start Google Chrome, Microsoft Edge or Mozilla Firefox and open the application link given under “Project at a Glance”.
2. **Sign in.** Log in with the Zoho account that has been given access to the application.
3. **Choose the application.** Open Personal Expense Tracker. The menu shows Dashboard, Expense Entries and Category Masters.
4. **Mobile access (optional).** Install the Zoho Creator app on Android or iOS, sign in with the same Zoho account and open Personal Expense Tracker from the application list.
5. **Sign out.** When finished, select the profile picture and choose Sign out.

D.2 Instructions for the Administrator

1. **Add a category.** Open Category Masters and select the add (+) control. Enter Category Name, Category Code, Color Code (for example #FF9800), Display Order and Total Allocated. Keep Is Active ticked and select Submit.
2. **Leave the balance fields to the workflow.** Current Spent, Remaining Balance and Last Updated are filled in automatically when expenses are submitted and need not be edited by hand.
3. **Change a budget or retire a category.** Open the record in the Category Masters report, change Total Allocated or untick Is Active, and save. The Remaining Balance is recalculated when the next expense for that category is submitted.
4. **Review spending.** Open the Dashboard to see Total Allocated, Total Spent, Remaining Balance, Total Transactions and the category-wise and monthly charts.
5. **Correct expense records.** Open Expense Entries, search for the record, then edit or delete it. The category balance is updated only when an expense is

created, so run the recalculation process (DS-02) after an expense is edited or deleted, or after earlier expenses are added.

6. **Check the email notification.** After each submitted expense, an email titled "New Expense Added – Personal Expense Tracker" should arrive in the inbox. If it does not, check the spam folder and the email notification settings of the Expense Entry workflow.
7. **Control access.** Share the application only with authorised users and give each user the minimum access needed. Take a backup before changing forms or workflows.

D.3 Instructions for the User

1. **Open the form.** Select Expense Entries and then the add (+) control to open the Expense Entry form.
2. **Enter the details.** Type the Expense Name (mandatory), choose the Expense Date, enter the Amount and add a Description if needed.
3. **Select the category.** Choose a Category Reference such as Food or Transport from the list.
4. **Submit.** Select Submit. The expense is saved and the category's Current Spent and Remaining Balance are updated automatically.
5. **Check the email.** A notification showing the Expense Name, Expense Date, Amount, Category and Description arrives in the inbox.
6. **View the history.** Open Expense Entries to see all saved expenses; use the search box to find a record by its category reference.
7. **Check budgets.** Open Category Masters to see the allocated, spent and remaining amounts, or open the Dashboard for the totals and charts.
8. **If a wrong entry is saved.** Edit or delete it if your access allows, and inform the Administrator so that the category balances can be recalculated.